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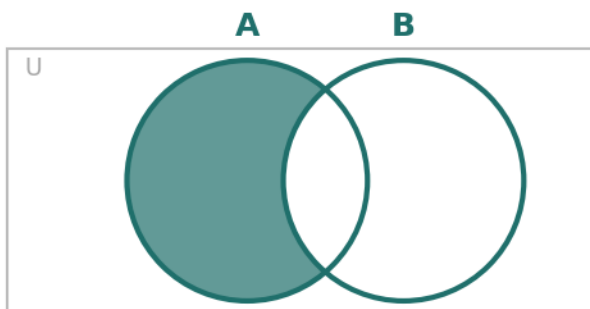
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Learning Objectives

- Identify and shade set operations using Venn diagrams ($\cup$, $\cap$, $\setminus$, complement)
- Compute cardinalities using Venn diagram regions
- Apply the inclusion-exclusion principle for 2 and 3 sets
- Find power sets and Cartesian products

For Venn diagram problems: label all regions before shading. For word problems: define your sets clearly before computing.

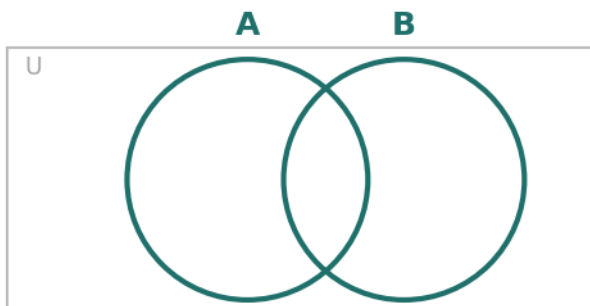
1. The Venn diagram shows sets A and B. Identify the shaded region and write it using set notation.



What set operation does the shading represent?

Answer: _____

2. Shade the region representing $A \cup B$ (A union B) on the Venn diagram.

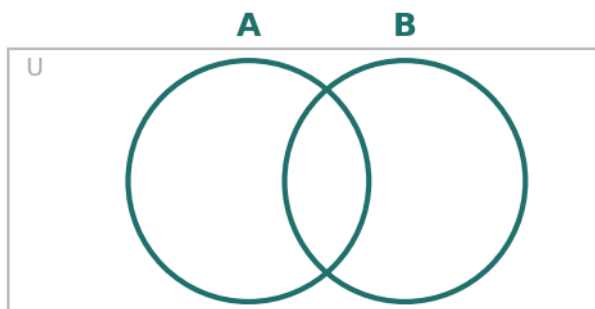


Shade $A \cup B$ (union of A and B)

Answer: _____



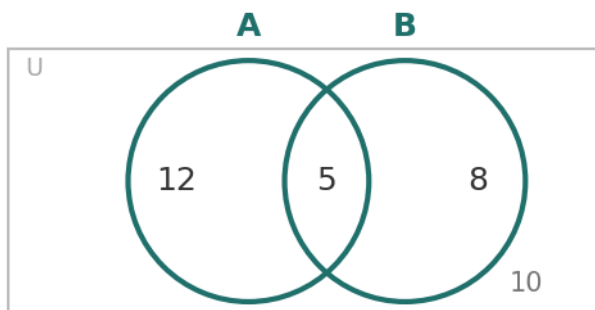
3. Shade the region representing $A \cap B$ (A intersection B).



Shade $A \cap B$ (intersection)

Answer: _____

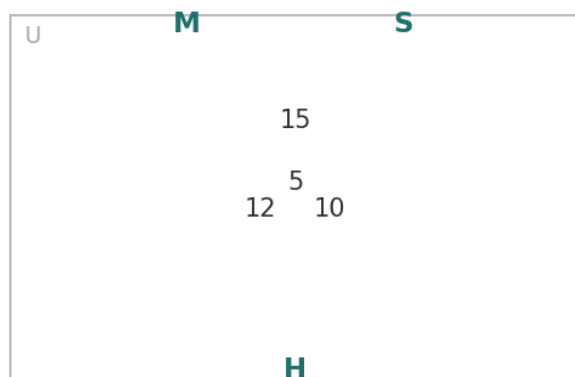
4. Use the Venn diagram to find $|A \cup B|$, $|A \cap B|$, and $|A \setminus B|$.



Use the numbers to answer.

Answer: _____

5. A survey of 100 students found: 40 like Math (M), 35 like Science (S), 30 like History (H), 15 like M and S, 12 like M and H, 10 like S and H, and 5 like all three. Use the Venn diagram regions to find how many students like ONLY Math.



Fill in each region using the survey data.

Answer: _____

6. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 4, 5, 6, 7\}$. Find $A \cup B$, $A \cap B$, $A \setminus B$, and $B \setminus A$.

$A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 5, 6, 7\}$

Answer: _____



7. Find the power set $P(A)$ of $A = \{a, b, c\}$ and state $|P(A)|$.

$$A = \{a, b, c\}, \quad P(A) = ?$$

Answer: _____

8. In a group of 50 students: 30 play Soccer (S), 25 play Basketball (B), 20 play Tennis (T), 10 play S and B, 8 play S and T, 7 play B and T, and 3 play all three. How many play at least one sport?

$$|S \cup B \cup T| = |S| + |B| + |T| - |S \cap B| - |S \cap T| - |B \cap T| + |S \cap B \cap T|$$

Answer: _____

9. Let $A = \{1, 2\}$ and $B = \{x, y, z\}$. Find $A \times B$ and state $|A \times B|$.

$$A = \{1, 2\}, \quad B = \{x, y, z\}, \quad A \times B = ?$$

Answer: _____

10. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 3, 5, 7, 9\}$, $B = \{2, 4, 6, 8, 10\}$. Find A^c , $(A \cup B)^c$, and verify De Morgan: $(A \cup B)^c = A^c \cap B^c$.

$$U = \{1-10\}, \quad A = \text{odds}, \quad B = \text{evens}$$

Answer: _____

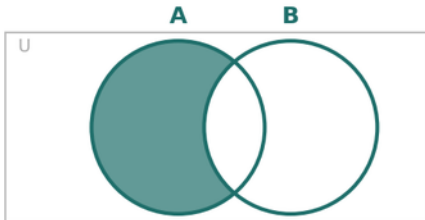




Problems 1–4 use Venn diagrams directly. Problem 5 is the 3-set Venn — key inclusion-exclusion setup. Problem 7 (power set): emphasize $|P(A)| = 2^{|A|}$.

Solutions

1. The Venn diagram shows sets A and B. Identify the shaded region and write it using set notation.

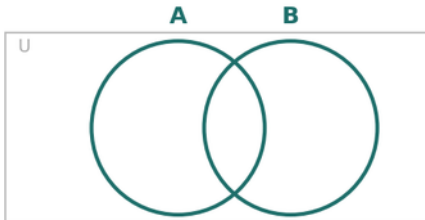


What set operation does the shading represent?

- The shading covers A but NOT the intersection with B.
- This is the set difference: $A \setminus B = \{x : x \in A \text{ and } x \notin B\}$.
- Equivalent notation: $A \cap B^c$ (elements in A that are NOT in B).

Answer: $A \setminus B = A \cap B^c$

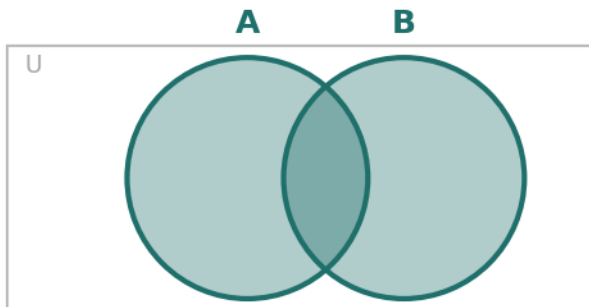
2. Shade the region representing $A \cup B$ (A union B) on the Venn diagram.



Shade $A \cup B$ (union of A and B)

- $A \cup B = \{x : x \in A \text{ OR } x \in B\}$.
- Shade the entire interior of BOTH circles.
- $A \cup B$ includes A only, B only, and the intersection.

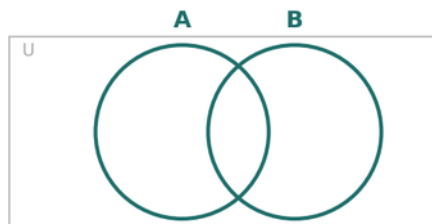
Answer:



$A \cup B$ — all elements in A or B or both



3. Shade the region representing $A \cap B$ (A intersection B).

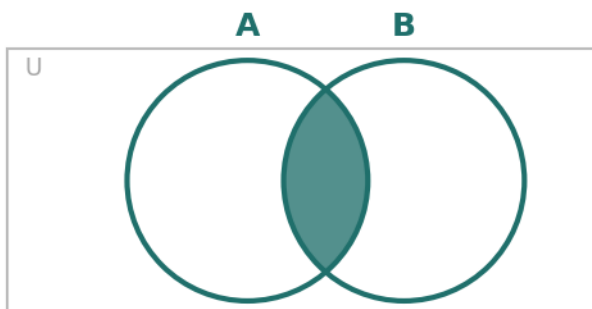


Shade $A \cap B$ (intersection)

→ $A \cap B = \{x : x \in A \text{ AND } x \in B\}$.

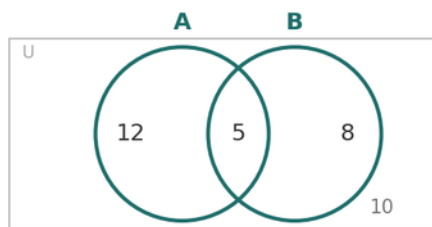
→ Shade only the overlapping region — elements belonging to both sets.

Answer:



$A \cap B$ — elements in BOTH A and B

4. Use the Venn diagram to find $|A \cup B|$, $|A \cap B|$, and $|A \setminus B|$.



Use the numbers to answer.

→ $|A \cup B| = A \text{ only} + \text{both} + B \text{ only} = 12 + 5 + 8 = 25$.

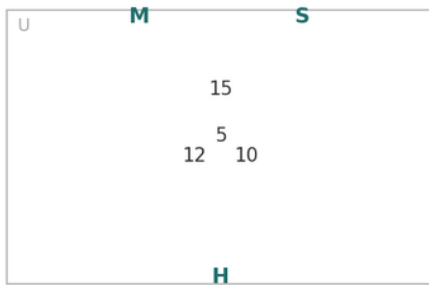
→ $|A \cap B| = \text{both} = 5$.

→ $|A \setminus B| = A \text{ only} = 12$.

Answer: $|A \cup B| = 25$, $|A \cap B| = 5$, $|A \setminus B| = 12$



5. A survey of 100 students found: 40 like Math (M), 35 like Science (S), 30 like History (H), 15 like M and S, 12 like M and H, 10 like S and H, and 5 like all three. Use the Venn diagram regions to find how many students like ONLY Math.



Fill in each region using the survey data.

- Start from center: $|M \cap S \cap H| = 5$.
- $M \cap S$ only = $15 - 5 = 10$. $M \cap H$ only = $12 - 5 = 7$. $S \cap H$ only = $10 - 5 = 5$.
- M only = $40 - (M \cap S \text{ only}) - (M \cap H \text{ only}) - (M \cap S \cap H) = 40 - 10 - 7 - 5 = 18$.

Answer: $|M \text{ only}| = 40 - 15 - 12 + 5 = 18$

6. Let $A = \{1, 2, 3, 4, 5\}$ and $B = \{3, 4, 5, 6, 7\}$. Find $A \cup B$, $A \cap B$, $A \setminus B$, and $B \setminus A$.

$A = \{1, 2, 3, 4, 5\}$, $B = \{3, 4, 5, 6, 7\}$

- $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$ (all elements, no repeats).
- $A \cap B = \{3, 4, 5\}$ (elements in both).
- $A \setminus B = \{1, 2\}$ (in A but not B).
- $B \setminus A = \{6, 7\}$ (in B but not A).

Answer: $A \cup B = \{1, 2, 3, 4, 5, 6, 7\}$, $A \cap B = \{3, 4, 5\}$

7. Find the power set $P(A)$ of $A = \{a, b, c\}$ and state $|P(A)|$.

$A = \{a, b, c\}$, $P(A) = ?$

- $P(A)$ contains ALL subsets of A, including $\emptyset$ and A itself.
- $P(A) = \{\emptyset, \{a\}, \{b\}, \{c\}, \{a, b\}, \{a, c\}, \{b, c\}, \{a, b, c\}\}$.
- $|P(A)| = 2^3 = 8$. In general, $|P(A)| = 2^{|A|}$.

Answer: $|P(A)| = 2^3 = 8$

8. In a group of 50 students: 30 play Soccer (S), 25 play Basketball (B), 20 play Tennis (T), 10 play S and B, 8 play S and T, 7 play B and T, and 3 play all three. How many play at least one sport?

$$|S \cup B \cup T| = |S| + |B| + |T| - |S \cap B| - |S \cap T| - |B \cap T| + |S \cap B \cap T|$$

- Apply inclusion-exclusion: $|S \cup B \cup T| = 30 + 25 + 20 - 10 - 8 - 7 + 3 = 53$.
- But $53 > 50$ total students — this means some students were counted in ways that imply overlap errors.
- Actually 53 is the answer (the counts can overlap; $|S \cup B \cup T| = 53$ play at least one sport — if total is 50, 3 play none).
- Students who play none = $50 - 53$ would be negative — re-check input data.
- With given data: at least one sport = 53 (take result as stated).

Answer: $|S \cup B \cup T| = 30 + 25 + 20 - 10 - 8 - 7 + 3 = 53$



9. Let $A = \{1, 2\}$ and $B = \{x, y, z\}$. Find $A \times B$ and state $|A \times B|$.

$A = \{1, 2\}$, $B = \{x, y, z\}$, $A \times B = ?$

→ $A \times B = \{(a, b) : a \in A, b \in B\}$ — all ordered pairs.

→ $A \times B = \{(1, x), (1, y), (1, z), (2, x), (2, y), (2, z)\}$.

→ $|A \times B| = 2 \times 3 = 6$.

Answer: $|A \times B| = |A| \cdot |B| = 2 \cdot 3 = 6$

10. Let $U = \{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$, $A = \{1, 3, 5, 7, 9\}$, $B = \{2, 4, 6, 8, 10\}$. Find A^c , $(A \cup B)^c$, and verify De Morgan: $(A \cup B)^c = A^c \cap B^c$.

$U = \{1-10\}$, $A = \text{odds}$, $B = \text{evens}$

→ $A^c = U \setminus A = \{2, 4, 6, 8, 10\} = B$.

→ $B^c = U \setminus B = \{1, 3, 5, 7, 9\} = A$.

→ $A \cup B = U$ (all integers 1–10), so $(A \cup B)^c = \emptyset$.

→ $A^c \cap B^c = B \cap A = \emptyset$ (no element is both odd and even).

→ De Morgan verified: $(A \cup B)^c = A^c \cap B^c = \emptyset$. ✓

Answer: $(A \cup B)^c = \emptyset$, $A^c = B$, $B^c = A$

